CISCO IOS COMMAND CHEATSHEET

Practical Shorthand Reference for Lab Work

Router • Switch • Routing • Switching • DHCP • VLAN

Shorthand used throughout: `en` = enable • `conf t` = configure terminal • `int gig0/0` = interface GigabitEthernet 0/0 • `ip add` = ip address • `no sh` = no shutdown • `sh` = show • `wr` = write memory (copy run start)

Note: All interface examples use **gig** (GigabitEthernet). Replace `0/0`, `0/1` etc. with the actual port numbers on your device. Commands are shown in the order you typically type them.

1. CLI Modes & Navigation

`en`	Enter privileged EXEC mode (from user mode)
`conf t`	Enter global configuration mode
`interface gig0/0`	Enter interface configuration mode (GigabitEthernet 0/0)
`exit`	Go back one level
`end or Ctrl+Z`	Return directly to privileged EXEC mode
`disable`	Return from privileged EXEC to user EXEC

2. Basic Device Configuration (Router / Switch)

`hostname R1`	Set device hostname (use R1, R2, S1, S2...)
`enable secret cisco`	Set encrypted privileged mode password (preferred)
`enable password cisco`	Set clear-text privileged password (legacy)
`service password-encryption`	Encrypt all plain-text passwords in config
`no ip domain-lookup`	Disable DNS lookup for mistyped commands (very useful)
`banner motd #Unauthorized!#`	Set message-of-the-day banner
`line console 0`	Enter console line configuration
`password cisco`	Set console password
`login`	Require login on console
`logging synchronous`	Prevent log messages from interrupting typing
`line vty 0 15`	Enter virtual terminal (Telnet/SSH) lines
`password cisco`	Set VTY password
`login`	Require login on VTY lines
`transport input all`	Allow Telnet + SSH (or 'ssh' only)

3. Interface Configuration (GigabitEthernet)

`int gig0/0`	Select GigabitEthernet 0/0
`ip add 192.168.1.1 255.255.255.0`	Assign IP address and subnet mask
`no sh`	Enable the interface (bring it up)
`description Link-to-R2`	Add a helpful description
`duplex auto`	Set duplex (auto / full / half)
`speed auto`	Set speed (auto / 10 / 100 / 1000)
`shutdown`	Administratively disable the interface
`no ip add`	Remove IP address from interface

Tip: On routers, always no sh after assigning IP. On switches, access ports are up by default once a cable is connected.

4. Essential Show / Verification Commands

`sh ip int br`	Interface status + IP summary
`sh run`	Running configuration
`sh start`	Startup configuration
`sh ip route`	Routing table
`sh protocols`	IP status of interfaces
`sh version`	IOS version & uptime
`sh cdp neighbors`	Directly connected Cisco devices
`sh cdp neighbors detail`	Detailed CDP info (IP, platform)

`sh vlan brief`	VLAN list + port membership
`sh int trunk`	Trunk interfaces status
`sh int switchport`	Switchport details
`sh ip dhcp binding`	DHCP client leases
`sh ip dhcp pool`	DHCP pool statistics
`sh ip ospf neighbor`	OSPF neighbors
`sh ip eigrp neighbors`	EIGRP neighbors
`sh ip protocols`	Active routing protocols

5. Saving & Managing Configuration

`wr or copy run start`	Save running-config to startup-config (NVRAM)
`copy start run`	Load startup-config into running-config
`erase startup-config`	Delete saved configuration (reload will wipe)
`reload`	Reboot the device
`show flash:`	View flash memory contents

6. DHCP Server Configuration (on Router)

Typical sequence on a router acting as DHCP server:

`ip dhcp excluded-address 192.168.1.1 192.168.1.10`	Exclude static addresses (router, servers, printers)
`ip dhcp pool LAN1`	Create a DHCP pool named LAN1
`network 192.168.1.0 255.255.255.0`	Network and mask for the pool
`default-router 192.168.1.1`	Default gateway given to clients
`dns-server 8.8.8.8 8.8.4.4`	DNS servers
`domain-name example.local`	Domain name (optional)
`lease 3`	Lease time in days (or 'infinite')
`exit`	Leave pool configuration

Verification: `sh ip dhcp binding` | `sh ip dhcp pool` | On client PC set IP Configuration → DHCP

DHCP Relay (helper-address): On the interface facing clients that are remote from the DHCP server → `ip helper-address 192.168.10.1`

7. Static Routing

`ip route 192.168.2.0 255.255.255.0 10.0.0.2`	Static route: network → next-hop IP
`ip route 192.168.2.0 255.255.255.0 gig0/1`	Static route using exit interface (point-to-point)
`ip route 0.0.0.0 0.0.0.0 10.0.0.1`	Default route (gateway of last resort)
`ip route 192.168.3.0 255.255.255.0 10.0.0.2 5`	Floating static (AD=5) for backup path
`no ip route 192.168.2.0 255.255.255.0 10.0.0.2`	Remove a static route

Verification: `sh ip route` (look for S code) | `sh ip route static` | ping / traceroute remote networks

8. RIPv2 Configuration

`router rip`	Enter RIP routing process
`version 2`	Use RIPv2 (classless, supports VLSM)
`no auto-summary`	Disable automatic classful summarization (important!)
`network 192.168.1.0`	Advertise network (classful statement – still works with VLSM)
`network 10.0.0.0`	Advertise another network
`passive-interface gig0/2`	Stop sending RIP updates out this interface
`exit`	Leave router configuration

Verification: `sh ip protocols` | `sh ip route` (code R) | `debug ip rip` (careful – noisy)

9. OSPF (Single-Area) Configuration

`router ospf 1`	Start OSPF process ID 1 (local significance)
`router-id 1.1.1.1`	Manually set Router ID (recommended)
`network 192.168.1.0 0.0.0.255 area 0`	Advertise network with wildcard mask → area 0
`network 10.0.0.0 0.0.0.3 area 0`	Example for a /30 link (wildcard = 0.0.0.3)
`passive-interface gig0/2`	Do not form adjacencies / send hellos on this interface
`auto-cost reference-bandwidth 1000`	Adjust cost calculation for Gigabit (optional)
`exit`	Leave OSPF configuration

Verification: `sh ip ospf neighbor` | `sh ip ospf int brief` | `sh ip route ospf` (code O) | `sh ip protocols`

Wildcard tip: Wildcard mask = inverse of subnet mask. /24 → 0.0.0.255 | /30 → 0.0.0.3 | /16 → 0.0.255.255

10. EIGRP Configuration

`router eigrp 100`	Start EIGRP with Autonomous System number 100 (must match)
`no auto-summary`	Disable automatic summarization (critical for VLSM)
`network 192.168.1.0`	Advertise network (classful style)
`network 10.0.0.0 0.0.0.3`	Or with wildcard for precision
`passive-interface gig0/2`	Suppress EIGRP hellos on this interface
`eigrp router-id 1.1.1.1`	Optional: set Router ID
`exit`	Leave EIGRP configuration

Verification: `sh ip eigrp neighbors | sh ip eigrp topology | sh ip route eigrp` (code D) `| sh ip protocols`

11. VLAN & Trunk Configuration (Switch)

Create VLANs

`vlan 10`	Create VLAN 10
`name SALES`	Assign a descriptive name
`vlan 20`	Create VLAN 20
`name ENGINEERING`	Name it
`vlan 99`	Create VLAN 99 (often used as native / management)
`name NATIVE`	Name it
`exit`	Return to global config

Access Ports

`int gig0/1`	Select access port
`switchport mode access`	Force access mode
`switchport access vlan 10`	Assign to VLAN 10
`no sh`	(usually already up)
`int range gig0/2 - 5`	Configure a range of ports at once
`switchport mode access`	Access mode
`switchport access vlan 20`	Assign to VLAN 20

Trunk Ports

`int gig0/24`	Select trunk port (usually uplink)
`switchport mode trunk`	Force trunk mode
`switchport trunk encapsulation dot1q`	(needed on some older platforms)
`switchport trunk native vlan 99`	Set native VLAN (untagged)
`switchport trunk allowed vlan 10,20,99`	Limit allowed VLANs (security best practice)
`no sh`	Ensure interface is up

Verification: `sh vlan brief | sh int trunk | sh int switchport`

12. Inter-VLAN Routing – Router-on-a-Stick

Connect router `gig0/0` to a switch trunk port. Create one subinterface per VLAN:

`int gig0/0`	Physical interface (no IP here)
`no sh`	Bring physical interface up
`int gig0/0.10`	Subinterface for VLAN 10
`encapsulation dot1Q 10`	Tag frames with VLAN 10
`ip add 192.168.10.1 255.255.255.0`	Gateway for VLAN 10
`int gig0/0.20`	Subinterface for VLAN 20
`encapsulation dot1Q 20`	Tag frames with VLAN 20
`ip add 192.168.20.1 255.255.255.0`	Gateway for VLAN 20
`int gig0/0.99`	Subinterface for native / management VLAN
`encapsulation dot1Q 99 native`	Native (untagged) – optional keyword
`ip add 192.168.99.1 255.255.255.0`	Management IP

Verification: From a PC in VLAN 10 ping gateway 192.168.10.1 and a PC in VLAN 20. On router: `sh ip int br` (should show subinterfaces up/up).

13. Quick Troubleshooting Checklist

Symptom	Common Causes & Commands to Check
Interface down / down	`sh ip int br` → missing `no sh`, wrong cable, speed/duplex mismatch
Interface up / down	Cable problem, remote side shutdown, or Layer-1 issue
No ping (same VLAN)	Wrong IP/mask, wrong VLAN assignment, port security, ACL
No inter-VLAN ping	Missing subinterface, wrong encapsulation, trunk not allowing VLAN, missing native VLAN match
Route missing	`sh ip route` → check network statements, passive-interface, AD, next-hop reachability
OSPF/EIGRP no neighbor	Mismatched AS/process, timers, authentication, passive-interface, different subnet, ACL blocking hellos
DHCP not working	Pool network mismatch, exclusions wrong, missing default-router, helper-address missing, interface down
Trunk not forming	`sh int trunk` → both sides must be trunk, allowed VLANs match, native VLAN consistent

14. One-Page Quick Sequence Examples

A. New Router – Basic Setup

```
en
conf t
hostname R1
no ip domain-lookup
enable secret cisco
line console 0
  password cisco
  login
  logging synchronous
line vty 0 15
  password cisco
  login
int gig0/0
  ip add 192.168.1.1 255.255.255.0
  no sh
  description LAN
int gig0/1
  ip add 10.0.0.1 255.255.255.252
  no sh
  description Link-to-R2
end
wr
```

B. Router-on-a-Stick + Two VLANs (quick)

```
en → conf t
int gig0/0 → no sh → exit
int gig0/0.10 → encapsulation dot1Q 10 → ip add 192.168.10.1 255.255.255.0
int gig0/0.20 → encapsulation dot1Q 20 → ip add 192.168.20.1 255.255.255.0
end → wr
```

C. OSPF Single Area (minimal)

```
en → conf t
router ospf 1
  router-id 1.1.1.1
  network 192.168.1.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.0.3 area 0
end → wr
```

Keep this cheatsheet open while working in Packet Tracer. Always verify with the appropriate show commands after every major change. Practice the sequences until they become muscle memory.

Cisco IOS Command Cheatsheet • GigabitEthernet Focus • Lab-Ready Shorthand